worldpay from FIS

FIS Zabbix (NFT) US Configuration

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Hosting Monitoring

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1. Document Updates

This section shows the versions of the document and the changes to the document.

Date	Version	Updated By	Details
24/05/2023	1.0	A Dance	Initial release
31/05/2023	1.1	A Dance	The Configure database down alert section has had a note added to state that the alert should not be enabled until advised to do so. The dependency is an update to the FIS-Netcool / FIS- Aureus interface to handle the alerts as state based with a timed delay. This requirement is to prevent false alerts from patching.
01/06/2023	1.2	A Dance	Added Solaris Autoregistration steps
01/06/2023	1.3	A Dance	Updated Autoregistration for TEST servers inclusion regex
01/06/2023	1.4	A Dance	Added section for Maintenance periods
05/06/2023	1.5	A Dance	Added new section : Configure Zabbix Agent Autoregistration Pre-Shared Key
08/06/2023	1.6	A Dance	Added corrections to the Solaris Autoregistration
14/06/2023	1.7	A Dance	Added certificate creation and implementation section Added tool variable Netcool interface reference table link
10/07/2023	1.8	A Dance	Updated Netcool interface configuration to include zabbix.conf file
01/09/2023	1.9	A Dance	Updated web server /zabbix/data to /zabbix/web
25/09/2023	2.0	A Dance	Added the Netcool Bi-Directional integration configuration account and token creation
12/10/2023	2.1	A Dance	Added Zabbix Performance Tuning section – added Proxy Tuning details
16/10/2023	2.2	A Dance	Update to proxy server tuning, added StartPingers=100 and StartHistoryPollers=20
2/1/2024	2.3	A Dance	Updated Auto Registration Actions to add leading “^” to exclude hosts with Z prefix from sending to netcool group
25/01/2024	2.4	A Dance	Added section to configure the postgres database monitoring Added section to tune the Apache php web server

14/02/2024	2.5	A Dance	Added new proxy configuration recommendations

2. Overview

The document provides the configuration steps required to Configure the FIS Zabbix monitoring infrastructure. The Configuration provides no fault tolerance in the server infrastructure, (NFT).

NOTES:

1. The configuration procedure does not support SE Linux
2. The configuration only supports Linux 8

The build process utilizes the following software revision components.

- FIS Banking Netcool Interface version 3.6
- FIS Zabbix – Aureus Interface version 1.7
- Zabbix version 6.0.18

3. Description of the service

The Zabbix infrastructure provides Operating system and Application monitoring facilities, utilizing Agent and non-agent technologies.

4. Engagement references

The following shows the list of Engagement requests for the service.

Service Request Date:	Reference Number:	Raised By:

5. Defined Service and Support Names

The Service Name: Zabbix Monitoring

The Support Group Name: Event Monitoring – Enterprise (SNOW Group)

6. Required Pre-requisites

The following will be required to implement the build procedure

1. root access to the servers in the infrastructure stack
2. access to the RPMs located in SharePoint (use the following link Netcool & Aureus)
3. The ability to copy the RPMs to the required destination server
4. The required password to apply to configure the LDAP interface in zabbix

7. LDAP Service account password details

The LDAP password details are available in the FIS CyberArk system.

To access the password details, follow the procedure below.

NOTE: Your “e” account will need access to the default tools Safe of (ETS-ETA_Tools-AppAdm)

1. Logon to a windows bastion server
5. Open an I.e. explorer browser, other browsers will have problems.
6. Navigate to: https://cyberark.fnfis.com/
7. Logon with your “e” account and password
8. In the search box (Top right) enter SvcAcctZabbixLDAP
9. Tick the box on the left of the entry you wish to access and click the entry once
10. On the box on the left, click the Copy button, Enter the Change, or Incident reference
11. The details are not in your clipboard, you may wish to paste it to a notepad

8. FIS Configuration sections

8.1. Overview

- The configuration procedure covers the required stages of configuration, it is suggested that the configuration is completed in the following order:

  - Zabbix Integration with LDAP
  - Zabbix Integration with netcool (for events)
  - Zabbix Integration with Aureus (sending realtime metrics and events to Aureus)
  - Importing Media Types for netcool and Aureus
  - Importing FIS Zabbix baseline templates

8.2. Creating base user groups

- The base user groups to be created are as follows

User Group Name	Frontend access	Enabled
General Users	LDAP	Ticked
internalAuthentication	Internal	Ticked

1. Select Administration / User groups
2. Select the Create user group button, top right
3. Enter the user group name from the table above
4. Select the Frontend access drop down selection from the table above
5. Check that the Enabled tick box is set from the table above
6. Select the Add button

Repeat for all the user groups in the table above

8.3. Create a Super Admin user

- The Super Admin user will be used for access to the Zabbix environment if the LDAP integration section below fails, this is to prevent being locked out of the zabbix instance.

1. Select, Administration, Users
2. Select Create user button, (top right)
3. Enter your “e” account number, e.g., e559158
4. Enter your Name and Last Name
5. In groups, select internalAuthentication
6. Enter a password for your “e” account
7. Enter the password (once again)
8. Select the Permissions TAB
9. In the Role field, select Super admin role
10. Select the Add button at the bottom of the screen

This account can now be used to login if the integration with LDAP has issues

8.4. Update Zabbix Severity settings

- This section will change the default zabbix severities to the normal FIS severities

1. Login to the Zabbix GUI with an account with super admin privileges
2. Select Administration, General, Trigger displaying options
3. Update the following fields

```
Disaster    =    Critical
High        =    Major
Average     =    Minor
Warning     =    Warning
```

NOTE: The entries are case sensitive

4. Select the Update button

8.5. Zabbix Integration with LDAP

The integration with LDAP allows the accounts in Zabbix to be authenticated against the FIS LDAP service.

8.5.1 Validate connectivity to the LDAP Service

The connectivity from the zabbix web server to the LDAP service should be checked before implementing LDAP in Zabbix

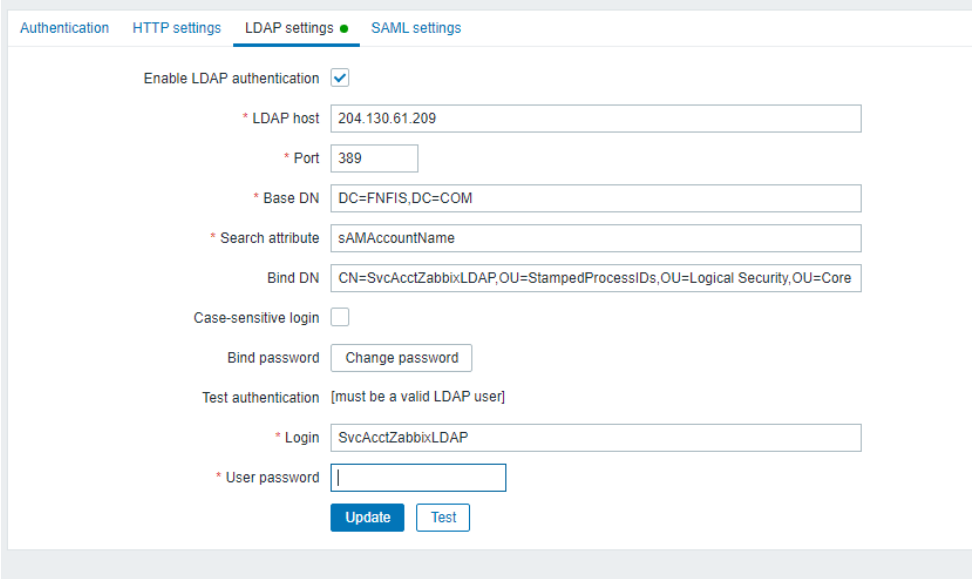
To validate connectivity, do the following:

1. Login to the zabbix web server
2. telnet 204.130.61.209 389
3. check telnet connects

8.5.2 Steps to integrate LDAP

1. Login to the Zabbix GUI and select the administration menu
2. Select the Authentication option
3. Select Default authentication: Internal
4. Select the LDAP settings TAB
5. Enter the following values for each option
   - Enable LDAP authentication: Ticked
   - LDAP host: 204.130.61.209
   - Port: 389
   - Base DN: DC=FNFIS,DC=COM
   - Search attribute: sAMAccountName
   - Bind DN: CN=SvcAcctZabbixLDAP,OU=StampedProcessIDs,OU=Logical Security,OU=Core Directory Services,DC=FNFIS,DC=com
   - Case-sensitive login Ticked
   - Bind password: {password for SvcAcctZabbixLDAP}
   - Login: SvcAcctZabbixLDAP

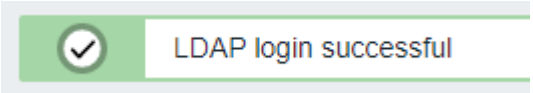
The LDAP settings screen should be showing the following



- User password: {password for SvcAcctZabbixLDAP}

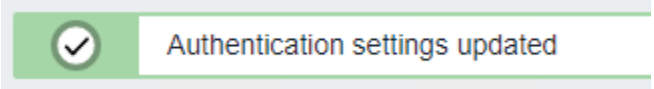
6. Select the Test button

The following should be shown:



7. Select the Update button

The following should be shown:



The LDAP Configuration has been completed.

8.6. Zabbix Netcool Banking interface Integration

The FIS netcool banking interface allows Zabbix to send events to the Netcool

8.6.1 Steps to install the Netcool Banking interface

1. Logon to the Zabbix Application server
2. Download the latest fis-zabbix-banking-netcool-x.x RPM file to the Zabbix Application server.
3. As root, install the RPM with the following command (example version 3.6 shown)

```
dnf install fis-zabbix-banking-netcool-3.6-1.el7_9.noarch.rpm
```

4. change the service working directory with the following commands (assuming /zabbix as the required filesystem)

```
cd /opt/netcool/setup
./remove
./install -d /zabbix
```

5. Update the netcool zabbix instance name

```
cd /opt/netcool/config
vi netcoolsvc.conf
update the line:
tool=zabbix_emea to tool=zabbix_{your instance name}
(Do not use spaces in the name)
     e.g., for voorhees use tool=zabbix_voorhees
The tool value reference for zabbix instances can be found here
```

6. Update the netcool zabbix instance name

```
cd /opt/netcool/config
vi zabbix.conf
update the line:
tool=zabbix_emea to tool=zabbix_{your instance name}
```

7. Update the netcool zabbix instance name

```
cd /opt/netcool/config
vi zabbix.conf
```

update the line:

FISDC=99 to the FISDC number that the Zabbix application server is located in. A list of FIS DC numbers can be located here

8. Restart the netcool interface service

```
systemctl restart fis-netcool
```

8.7. Zabbix aureus interface Integration

The FIS Aureus interface allows Zabbix to send events to the Aureus platform

8.7.1 Steps to install the Aureus interface

1. Logon to the Zabbix Application server
2. Download the latest fis-zabbix-aureus-x.x RPM file to the Zabbix Application server.
3. As root, install the RPM with the following command (example version 1.7 shown)

```
dnf install fis-zabbix-aureus-1.7-1.el7_9.noarch.rpm
```

4. change the service working directory with the following commands (assuming /zabbix as the required filesystem)

```
cd /opt/aureus/setup
./remove
./install -d /zabbix
```

5. Update the aureus zabbix exportdata directory value to the required filesystem path

```
cd /opt/aureus/config
vi aureussvc.conf
update the line:
metricexportpath=/zabbix/exportdata
NOTE: this assumes the zabbix server has been configured to
export data to the /zabbix/exportdata directory
i.e., the zabbix_server.conf value of the line Exportdir
(Do not use spaces in the path)
```

6. Restart the netcool interface service

```
systemctl restart fis-aureus
```

8.8. Import Media Types, create users and groups for Netcool and Aureus

The media Types are imported in the Zabbix GUI to enable Zabbix to send events to the Netcool and Aureus interfaces.

8.8.1 Steps to Import the media types

1. The Media Type yaml files are located at this link
2. Download the media type yaml files to your local laptop or where you are using the Zabbix GUI
3. Login to the Zabbix GUI
4. Navigate to Administration / Media Types
5. Select the Import button, top right
6. Select Choose file, select one of the yaml files
7. If the Media Type does not exist in the Zabbix instance, select the Create new tick box
8. Select the Import button
9. The Media type will be imported
10. Repeat the above steps for the other yaml file, from step 6.

8.8.2 Steps to create a Notifications role

1. Login to the Zabbix GUI
2. Navigate to Administration / User roles
3. Select Create user role, top right
4. Update the following

Name	=	"Notifications"
User Type	=	User
Monitoring	=	Dashboard, Problems, Hosts, Latest data, Maps
Services	=	Services
Inventory	=	Overview, Hosts
Reports	=	Availability report, Triggers top 100

Default access to new UI elements	ticked
Default access to modules	ticked
Access to API	enabled

API methods = Deny list

Access to actions = Create and edit dashboards,
Create and edit maps
add problem comments
Change severity
Acknowledge problems
Close problems
Execute scripts
Manage API Tokens

Default access to new actions ticked

5. Select the Add button

8.8.3 Steps for the creation of user accounts for the netcool interface

1. Select the menu option Administration / Users
2. **Create a new user account for the netcool interface**

Username	=	`Netcool prod`
Name	=	`FIS Netcool`
Last name	=	`prod`
Groups	=	internalAuthentication
Password	=	{create a random password}
Time zone	=	System default

3. Select the Media TAB
4. Select the Add link
5. For Type, Select FIS-Netcool
6. In Send To, enter prod
7. Select the Add button
8. Select the Permissions TAB
9. Select the Role, Notifications
10. Select the Add button

1. Select the menu option Administration / Users
2. **Create a new user account for the aureus interface**

Username	=	`Aureus`
Name	=	`Aureus`
Last name	=	`prod`
Groups	=	internalAuthentication
Password	=	{create a random password}
Time zone	=	(UTC+00:00) UTC (This MUST be UTC)

3. Select the Media TAB
4. Select the Add link
5. For Type, Select FIS-Aureus
6. In Send To, enter prod
7. Select the Add button
8. Select the Permissions TAB
9. Select the Role, Notifications
10. Select the Add button

8.8.4 Steps to Create new Host and User group for the Netcool interface

1. Login to the Zabbix GUI
2. Navigate to Configuration / Host Groups
3. Select the Create host group, Top right
4. Enter the Group Name of: `Send to FIS-Netcool Prod`
5. Select the Add button
6. Navigate to Administration / User groups
7. Select the Create user group, Top right
8. Enter the group name of: `Send to FIS-Netcool Prod`
9. In the Users field, select: `Netcool prod`
10. For the Frontend access, select: `Internal`

11. Select the Permissions TAB
12. Select the following with Read access: `Send to FIS-Netcool Prod`
13. Select the Add link
14. Select the Add Button

8.8.5 Steps to Create new Host and User group for the Aureus interface

1. Login to the Zabbix GUI
2. Navigate to Configuration / Host Groups
3. Select the Create host group, Top right
4. Enter the Group Name of: `Send to FIS-Aureus Prod`
5. Select the Add button
6. Navigate to Administration / User groups
7. Select the Create user group, Top right
8. Enter the group name of: `Send to FIS-Aureus Prod`
9. In the Users field, select: `Aureus`
10. For the Frontend access, select: `Internal`
11. Select the Permissions TAB
12. Select the following with Read access: `Send to FIS-Aureus Prod`
13. Select the Add link
14. Select the Add Button

8.8.6 Steps to create Trigger action for the FIS-Netcool interface

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Trigger actions
3. Select Create action button, top right
4. Enter the Name: `Send to FIS-Netcool prod`
5. Select the Add link, in Conditions
6. Select Type: `Host Group`
7. Select the Operator `equals`
8. Select the Host Group of: `Send to FIS-Netcool Prod`
9. Select the Add button
10. Check the Enabled button is Ticked
11. Select the Operations TAB
12. In the Operations box, select the Add link
13. In the Send to users box, select the Add link
14. Tick the User: `Netcool prod`
15. Select the Select Button
16. Select the Add Button
17. In the Recovery Operations box, select the Add link
18. In the Send to users box, select the Add link
19. Tick the User: `Netcool prod`
20. Select the Select Button
21. Select the Add Button
22. In the Update Operations box, select the Add link
23. In the Send to users box, select the Add link
24. Tick the User: `Netcool prod`
25. Select the Select Button
26. Select the Add Button
27. Select the Add Button

8.8.7 Steps to create Trigger action for the FIS-Aureus interface

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Trigger actions
3. Select Create action button, top right
4. Enter the Name: `Send to FIS-Aureus prod`
5. Select the Add link, in Conditions
6. Select Type: `Host Group`
7. Select the Operator `equals`
8. Select the Host Group of: `Send to FIS-Aureus Prod`
9. Select the Add button
10. Check the Enabled button is Ticked
11. Select the Operations TAB
12. In the Operations box, select the Add link
13. In the Send to users box, select the Add link
14. Tick the User: `Aureus`
15. Select the Select Button
16. Select the Add Button
17. In the Recovery Operations box, select the Add link
18. In the Send to users box, select the Add link
19. Tick the User: `Aureus`
20. Select the Select Button
21. Select the Add Button
22. In the Update Operations box, select the Add link
23. In the Send to users box, select the Add link
24. Tick the User: `Aureus`
25. Select the Select Button
26. Select the Add Button
27. Select the Add Button

8.9. Create new FIS Host groups

The list of hosts groups will be required to be added for the steps further on in the document

8.9.1 Steps to create new host group

1. Login to the Zabbix GUI
2. Navigate to Configuration / Host groups
3. Select Create host group button, top right
4. Create the host groups as shown in the table below
5. Select the Update button to save the new Group

Host Group name
FIS AIX
FIS Linux
FIS Solaris
FIS Windows
FIS Solaris
FIS - do not send to netcool (servers in maintenance)
Pending decommission
TEST
TEST/AIX
TEST/Linux
TEST/Windows
TEST/Solaris

8.10. Create new FIS Host Template groups

The list of hosts template groups will be required to be added for the steps further on in the document

8.10.1 Steps to create new host Template groups

1. Login to the Zabbix GUI
2. Navigate to Configuration / Host groups
3. Select Create host group button, top right
4. Create the host groups as shown in the table below
5. Select the Update button to save the new Group

Host Template Group name	
Templates/FIS/AIX/Custom	Templates/FIS/Solaris/Custom
Templates/FIS/AIX/Override	Templates/FIS/Solaris/Override
Templates/FIS/Linux/Custom	Templates/FIS/Windows/Custom
Templates/FIS/Linux/Override	Templates/FIS/Windows/Override
Templates/FIS	

8.11. Import Zabbix base Templates

The zabbix base templates are used to monitor base OS metrics and raise alerts / problems

8.11.1 Steps to import the base templates

1. The template yaml files are numbered and are located at this link
2. Download the Template yaml files to your local laptop or where you are using the Zabbix GUI
3. Login to the Zabbix GUI
4. Navigate to Configuration / Templates
5. Select the Import button, top right

   NOTE: The templates need to be imported in order, i.e., template 1 first, then 2,3,4 etc
6. Select Choose file, select each one of the yaml files
7. If the template does not exist in the Zabbix instance, select the Create new tick box
8. Select the Import button
9. When the green template screen is seen, select the import button.
10. Repeat the import of the templates in sequence

8.12. Configure Auto Registration Actions

The auto registration actions will place hosts with a new installation of zabbix agent into the appropriate groups and assign the required templates for the operating system types

8.12.1 Steps to Add AIX Autoregistration

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions
3. Select Create action button, top right
4. In the Name field, enter “AIX Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “AIX”
9. Select the Add button
10. Click the Add link
11. Select Type “Host name”
12. Select Operator “does not match”
13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button
15. Click the Add link
16. Select Type “Host name”
17. Select Operator “does not match”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ule|vor|hop).**
19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “Send to FIS-Netcool Prod” and “Send to FIS-Aureus Prod”
24. Select the Add button
25. Click the Add link

26. Select Operation “Add to Host group”
27. In the Host Group field, enter “FIS AIX”
28. Select the Add button
29. Click the Add link
30. Select Operation “Link to template”
31. In the Templates field, enter “FIS-Global-AIX STD container”
32. Select the Add button
33. Click the Add link
34. Select Operation “Set host inventory mode”
35. In the Inventory mode field, select “Automatic”
36. Select the Add button
37. Select the Update button
38. If required, Select the Status Disabled to Enable the Autoregistration

8.12.2 Steps to Add AIX test servers

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions
3. Select Create action button, top right
4. In the Name field, enter “AIX test servers Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “AIX”
9. Select the Add button
10. Click the Add link
11. Select Type “Host name”
12. Select Operator “matches”
13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button

15. Click the Add link
16. Select Type “Host name”
17. Select Operator “matches”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ule|vor|hop).**
19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “TEST/AIX”
24. Select the Add button
25. Click the Add link
26. Select Operation “Link to template”
27. In the Templates field, enter “FIS-Global-AIX STD container”
28. Select the Add button
29. Select the Add button

8.12.3 Steps to Add Linux Autoregistration

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions
3. Select Create action button, top right
4. In the Name field, enter “Linux Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “Linux”
9. Select the Add button
10. Click the Add link
11. Select Type “Host name”
12. Select Operator “does not match”

13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button
15. Click the Add link
16. Select Type “Host name”
17. Select Operator “does not match”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ulw|vor|hop).**
19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “Send to FIS-Netcool Prod” and “Send to FIS-Aureus Prod”
24. Select the Add button
25. Click the Add link
26. Select Operation “Add to Host group”
27. In the Host Group field, enter “FIS Linux”
28. Select the Add button
29. Click the Add link
30. Select Operation “Link to template”
31. In the Templates field, enter “FIS-Global-Linux STD container”
32. Select the Add button
33. Click the Add link
34. Select Operation “Set host inventory mode”
35. In the Inventory mode field, select “Automatic”
36. Select the Add button
37. Select the Add button

8.12.4 Steps to Add Linux test servers

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions

3. Select Create action button, top right
4. In the Name field, enter “Linux test servers Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “Linux”
9. Select the Add button
10. Select the Add link
11. Select Type “Host name”
12. Select Operator “matches”
13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button
15. Click the Add link
16. Select Type “Host name”
17. Select Operator “matches”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ule|vor|hop).**
19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “TEST/Linux”
24. Select the Add button
25. Select the Add link
26. Select Operation “Link to template”
27. In the Templates field, enter “FIS-Global-Linux STD container”
28. Select the Add button
29. Select the Add button

8.12.5 Steps to Add Windows Autoregistration

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions
3. Select Create action button, top right
4. In the Name field, enter “Windows Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “Windows”
9. Select the Add button
10. Click the Add link
11. Select Type “Host name”
12. Select Operator “does not match”
13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button
15. Click the Add link
16. Select Type “Host name”
17. Select Operator “does not match”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ule|vor|hop).**
19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “Send to FIS-Netcool Prod” and “Send to FIS-Aureus Prod”
24. Select the Add button
25. Click the Add link
26. Select Operation “Add to Host group”
27. In the Host Group field, enter “FIS Windows”
28. Select the Add button
29. Select the Add link
30. Select Operation “Link to template”
31. In the Templates field, enter “FIS-Global-Windows STD container”

32. Select the Add button
33. Select the Add link
34. Select Operation “Set host inventory mode”
35. In the Inventory mode field, select “Automatic”
36. Select the Add button
37. Select the Add button

8.12.6 Steps to Add Windows test servers

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions
3. Select Create action button, top right
4. In the Name field, enter “Windows test servers Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “Windows”
9. Select the Add button
10. Select the Add link
11. Select Type “Host name”
12. Select Operator “matches”
13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button
15. Click the Add link
16. Select Type “Host name”
17. Select Operator “matches”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ule|vor|hop).**
19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “TEST/Windows”
24. Select the Add button
25. Select the Add link
26. Select Operation “Link to template”
27. In the Templates field, enter “FIS-Global-Windows STD container”
28. Select the Add button

29. Select the Add button

8.12.7 Steps to Add Solaris Autoregistration

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions
3. Select Create action button, top right
4. In the Name field, enter “Solaris Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “Solaris”
9. Select the Add button
10. Click the Add link
11. Select Type “Host name”
12. Select Operator “does not match”
13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button
15. Click the Add link
16. Select Type “Host name”
17. Select Operator “does not match”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ule|vor|hop).**
19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “Send to FIS-Netcool Prod” and “Send to FIS-Aureus Prod”
24. Select the Add button
25. Click the Add link
26. Select Operation “Add to Host group”
27. In the Host Group field, enter “FIS Solaris”
28. Select the Add button

29. Select the Add link
30. Select Operation “Link to template”
31. In the Templates field, enter “FIS-Global-Solaris STD container”
32. Select the Add button
33. Select the Add link
34. Select Operation “Set host inventory mode”
35. In the Inventory mode field, select “Automatic”
36. Select the Add button
37. Select the Add button

8.12.8 Steps to Add Solaris test servers

1. Login to the Zabbix GUI
2. Navigate to Configuration / Actions / Autoregistration actions
3. Select Create action button, top right
4. In the Name field, enter “Solaris test servers Autoregistration”
5. Click the Add link
6. Select Type “Host Metadata”
7. Select Operator “contains”
8. Enter the value “Solaris”
9. Select the Add button
10. Select the Add link
11. Select Type “Host name”
12. Select Operator “matches”
13. Enter the value: *^(zule4|zulw4|zbdc4|zlrk4|zpdc4|zvor4|z4w).**
14. Select the Add button
15. Click the Add link
16. Select Type “Host name”
17. Select Operator “matches”
18. Enter the value: *^(Z|z)(BDC|LRK|PDC|ULE|ULW|VOR|HOP|bdc|lrk|pdc|ule|ule|vor|hop).**

19. Select the Add button
20. Select the Operations Tab
21. Click the Add link
22. Select Operation “Add to Host group”
23. In the Host Group field, enter “TEST/Solaris”
24. Select the Add button
25. Select the Add link
26. Select Operation “Link to template”
27. In the Templates field, enter “FIS-Global-Solaris STD container”
28. Select the Add button
29. Select the Add button

The Autoregistration configuration is now complete

8.13. Configure Zabbix Agent Autoregistration Pre-Shared Key

The Zabbix Agent auto registration expects a known Zabbix agent ID and pre-shred key string, to allow the agent to register on the zabbix infrastructure. The following procedure will set the expected ID and Pre-Shared Key string

8.13.1 Steps to Configure the Zabbix Agent PSK

1. Login to the Zabbix GUI
2. Navigate to Administration / General / Autoregistration
3. On Encryption level, Untick: “No encryption” and Tick “PSK”
4. In the PSK Identity field, enter: “FIS AProxy” , NOTE: there is a space between “FIS and AProxy”
5. Retrieve the username “FIS (US/EMEA) Zabbix-agent-psk-id” from CyberArch “password” and enter it in the PSK Field

   NOTE:

   See section 7 for details on how to connect to CyberArch and retrieve the “password, which is the PSK”

6. Select the Update Button

   The Configuring the Zabbix Agent Autoregistration Pre-Shared Key is Complete

8.14. Configure Database down alert

The database down alert is used to send an alert if the Master Zabbix database is unavailable

8.14.1 Steps to Enable the Zabbix Master database down Alert

1. Login to the Zabbix GUI
2. Navigate to Administration / General / Other
3. Select User group for database down message and select "Send to FIS-Netcool Prod"
4. Select the Update Button

NOTE: As of 31St May 2023, the User group for database down message should not be BLANK. This is due to a requirement for the update of the FIS-Netcool interface to handle the Database down alert as a state-based alert. An advisory will be sent when the database down message can be enabled.

8.15. Configure retention settings

8.15.1 Steps to set data retention house keeping

The FIS standard data retention for Zabbix is 91 days of trend data and 3 days of history

1. Login to the Zabbix GUI
2. Navigate to Administration / General / Housekeeping
3. Set the following:

History (Data storage period)	3d
Trends (Data storage period)	91d

4. Select the Update button

8.16. Configure Maintenance settings

8.16.1 Steps to set Maintenance entries

1. Login to the Zabbix GUI
2. Navigate to Configuration / Maintenance
3. Select Create maintenance period button, top right
4. In the Name field, enter “pending decommission servers”
5. Select Maintenance type “No data collection”
6. In Active since, select the current date and time
7. In the Active till field, change the year to 2030
8. Periods, select the add link
9. In Period type, select “Daily”,
10. In Every days(s), select “1”
11. In At (hour:minute) select “00:00
12. In Maintenance period length, set Days to “1”
13. Select the Add button
14. In host groups, select “Pending decommission”
15. In the description field, enter “used to stop data collection and alerting for server that have been set as Pending decommission in the CMDB”
16. Select the Add button
17. Select Create maintenance period button, top right
18. In the Name field, enter “unmonitored servers”
19. Select Maintenance type “No data collection”
20. In Active since, select the current date and time
21. In the Active till field, change the year to 2030
22. Periods, select the add link
23. In Period type, select “Daily”,
24. In Every days(s), select “1”
25. In At (hour:minute) select “00:00
26. In Maintenance period length, set Days to “1”
27. Select the Add button
28. In host groups, select “FIS – do not send to netcool (servers in maintenance)”

29. In the description field, enter "Used for unmonitored servers, no data collection and no alerting to netcool"
30. Select the Add button

8.17. Setup Master database dumps – for backups

This section has been delayed; the next release of the configuration document will include the backup section.

8.18. Configure SSL Certificate for Zabbix

The pre-requisite for this section is that the fqdn for the service needs to be available,

e.g., https://zabbix.voorhees.fnfis.com

NOTE: This section should not be used as of 15/06/2023, until the DNS fqdn records are in place

8.18.1 Steps to implement a certificate

1. Generate a CSR
2. ssh to the zabbix web server
3. Run the following commands
4. openssl req -newkey rsa:2048 -keyout zabbix.key -out zabbix.csr
5. enter a passphrase "zabbix"
6. Us the following table as an example of what values to use in the openssl request

Country Name:	us
State or Province Name:	voorhees
Locality Name:	voorhees
Organization Name:	FIS
Organizational Unit Name:	CIO

Common name:	*zabbix-voorhees.fnfis.com*
Email Address:	Leave it blank by hitting Enter.
A challenge password:	Leave it blank by hitting Enter.
An optional company name:	Leave it blank by hitting Enter.

7. A zabbix.csr file and zabbix.key file will be created.
8. Login to keyfactor
9. Select enrolment.
10. Select template “Web Server (13 Month – client and Server auth) PROD
11. Select Certificate Authority “FISGLOBAL RSA PROD SUB CA2”
12. Copy and Paste the content of the zabbix.csr file into the CSR Content field
13. Select Device Type “Others”
14. Update the Description field with the IP address, fqdn of the host where the certificate will be installed and port number,
    e.g., 100.73.132.102 vor1rciozabwb01.FNFIS.com 443

15. In the Folder field select FIS Event Monitoring – Enterprise – PROD
16. In the Sub-Folder, Enter “Zabbix”
17. in the field “Application-Contact-Primary” Enter “Richard Derr”
18. in the field “Application-Contact-Secondary” enter “Andrew Dance”
19. Check the Certificate format is Base-64 encoded
20. Select the Enroll button
21. You will receive an email advising that the certificate has been generated
22. In “My Certificates” in Keyfactor
23. Select the required certificate
24. Select Download
25. Select format of PEM
26. Select the DOWNLOAD button
27. You now have the certificate.pem
28. Rename the certificate.pem, e.g. zabbix-voorheesfnfiscom.pem to zabbix.pem
29. rename the zabbix.key to zabbix.key.orig
30. Remove the passphrase from the zabbix.key with openssl rsa -in zabbix.key.orig -out zabbix.key
31. Enter the passphrase of zabbix
32. Copy the pem file to a zabbix web host, place it in a personal cert directory

33. Switch to root
34. Make a backup of /zabbix/web/conf/certs files
35. Copy the new zabbix.pem and zabbix.key files to /zabbix/web/conf/certs
36. Restart the php-fpm and httpd service

8.19. Configure Netcool Bi-Directional integration

The netcool Bi-directional integration allows the netcool impact manager to connect to the zabbix instance, web api, using a token for authentication. The impact manager will be able to auto close the problem in zabbix when a SNOW incident is resolved. If the issue has not actually been resolved zabbix will raise another problem and associated SNOW incident.

There are three parts to the confirmation in zabbix, these are as follows:

- Role
- Local Account
- User Group
- Token

The token will need to be put into the impact netcool impact manager rules to allow it to close problems in the zabbix instance. The procedure for netcool is not covered in this document.

8.19.1 Steps to create a role

1. Login to the Zabbix GUI
2. Navigate to Administration / User roles
3. Select Create user role, top right
4. Update the following

Name	=	"api-event-acknowledge"
User Type	=	User
Monitoring	=	Problems
Services	=	none
Inventory	=	none
Reports	=	none

Default access to new UI elements		not ticked
Read-Write access to services =		None
Read-only access to services	=	None
Default access to modules	=	not ticked
Access to API	=	enabled
API methods	=	Allow list
API methods	=	event.acknowledge
Access to actions	=	Add problem comments
		Acknowledge problems
		Close problems
Default access to new actions		not ticked

5. Select the Add button

8.20.1 Steps for the creation of the user account

1. Select the menu option Administration / Users
2. **Create a new user account for the netcool interface**

Username	=	api-event-acknowledge
Name	=	`api`
Last name	=	`event-acknowledge`
Groups	=	internalAuthentication
Password	=	{create a random password}
Time zone	=	System default

3. Select the Permissions TAB
4. Select the Role, api-event-acknowledge
5. Select the Add button

8.21.1 Steps for the creation of user group

1. Select the menu option Administration / User Groups
2. Create a new user group “api-event-acknowledge”
3. Select Users “api-event-acknowledge”
4. Frontend access “Internal”
5. Enabled
6. Select Permissions Tab
7. Select “Send to FIS-Netcool Prod” Read-write
8. Select the Add link
9. Select the Add button

8.22.1 Steps for the creation of an API token

1. Select the menu option Administration / General
2. Select API tokens

3. Select the button “Create API token” top right
4. Enter the name “api-event-acknowledge”
5. Select the User “api-event-acknowledge”
6. Enter in the description “Token for Netcool event acknowledge and close problems”
7. Untick expiration date and time
8. Check Enabled is ticked
9. Select the Add button
10. Select the copy to clipboard link
11. Paste the clipboard to a notepad and save the file as the zabbix instance event-acknowledge token”

    NOTE: Once the close button is selected, it is not possible to view the token
12. Select the close button

9. Zabbix Performance Tuning

9.1. Zabbix Server Tuning

- none

9.2. Zabbix Proxy Tuning

The following are guides for approx. 1000 attached Zabbix agents to a proxy host

StartPollers = 100

StartPreprocessors = 20

StartPollersUnreachable = 50

StartPingers=100

StartHistoryPollers=20

StartTrappers=25

ListenBacklog=512

9.3. Zabbix Web Server Tuning

If there are errors in the /var/log/httpd/serror_log file for memory allocation, update the following.

File location: /etc/php-fpm.d/zabbix.conf

Update: php_value[memory_limit] from 128M to 4G

Update: php_value[upload_max_filesize] = 2M to 8M

To apply the configuration run the following

suexec systemctl restart php-fpm httpd

If there are timeout issues with the zabbix API, update the following file and value

File = /etc/httpd/conf.d/zabbix.conf

Value = timeout=300 update to 1200

10. Zabbix Postgres database monitoring

10.1. Configuration

- Configure the zabbix account to have a path to the postgres binaries

  Check the zabbix account on the postgres server has a home directory

  Create a .bash_profile file in the home directory

  Add the following in the file (the example is for postgres 14, you may have postgres 15 installed)

```
##########################################
# Bash profile for Zabbix Prostgres DB #
##########################################
```

```
export PATH=$PATH:/usr/pgsql-14/bin
```

- Verify that there is an account zbx_monitor in the postgres database

```
Login as postgres:

    su –  postgres

    psql --dbname=zabbix

    list user accounts is:  \du
```

- If the zbx_monitor account is not shown, run the following command to create the account.

create role zbx_monitor with encrypted password '*{password}*';

ALTER ROLE "zbx_monitor" WITH LOGIN;

- Change the zbx_monitor password as required

```
su - postgres

psql

ALTER USER zbx_monitor WITH PASSWORD '{password}';
```

- Copy the user parameter file template_db_postgresql.conf to the /etc/zabbix/zabbix_agentd.d
- Restart the zabbix-agent service with (systemctl restart zabbix-agent)

Update the pg_hba.conf file

Add the line

```
host    all            zbx_monitor     localhost              trust
```